



The effect of Porter's competitive forces on competitive advantage and organisational performance and the moderating role of management accounting practices

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Abstract

This study examines the effect of the intensity of competitive forces on organisational performance and competitive advantage and the moderating role of the use of traditional and contemporary management accounting practices on this relationship. Using survey data collected from 505 US-based financial managers, the findings indicate that the use of traditional management accounting practices was found to positively moderate the effect of the intensity of competitive forces on competitive advantage. Additional exploratory analysis revealed that the extent of use of contemporary management accounting practices positively moderates the effect of the intensity of competitive forces on organisational performance in a product differentiation/high-force environment. The extent of use of contemporary management accounting practices was also found to positively moderate the effect of the intensity of competitive forces on competitive advantage in a low-cost/high-force. However, the extent of use of contemporary management accounting practices was found to negatively moderate the effect of the intensity of competitive forces on competitive advantage in a product differentiation/low-force environment. Finally, the extent of use of traditional management accounting practices was found to negatively moderate the effect of the intensity of competitive forces on organisational performance in the product differentiation/high-force model. These findings provide a preliminary insight into the important role of management accounting practices in the management of competitive forces.

Keywords Traditional · Contemporary · Management accounting practices · Competitive forces · Competitive advantage · Organisational performance

1 Introduction

Michael Porter's (1979) 'How competitive forces shape strategy' has influenced the field of research on competitive forces and the formation and effectiveness of strategies within organisations. Porter (1979) famously claimed that five forces shape the competitive environment of an organisation: the threat of new entrants, the bargaining power of suppliers, the bargaining power of buyers, rivalry among existing competitors, and the threat of substitute products or services. These five forces combine to shape the industry structure and influence the way organisations compete within an industry, with organisations tending to emphasise Porter's (1985) cost focus (i.e. low-cost strategy) or differentiation focus (i.e. product-differentiation strategy) generic competitive strategies.

It is important to consider the collective strength of the five forces (Ho, 2016; Zhao et al., 2016) with Porter (2008, p. 86) stating that "the strength of the five competitive forces determines the industry's long-run profit potential because it determines how the economic value created by the industry is divided—how much is retained by companies in the industry versus bargained away by customers and suppliers, limited by substitutes, or constrained by potential new entrants". Accordingly, many studies have focused on Porter's (1979) Five Forces Model to evaluate the competitive structures within specific industries (e.g. Carle et al., 2005; Rajasekar & Al Raei, 2013; Zhao et al., 2016) and to assess the competitiveness and profitability of specific industries and/or organisations. However, this research has tended to be qualitative in nature and emphasised case study analysis, with the limited empirical studies which have considered the effect of Porter's competitive forces on organisational outcomes (Barba-Sánchez et al., 2018) finding that stronger competitive forces inhibit corporate performance.

The first objective of this study is to contribute to this literature by examining the association between the collective strength of the five competitive forces, referred to in this study as the intensity of competitive forces, and the two organisational outcomes commonly associated with Porter's competitive forces: organisational performance and competitive advantage (Marin-Cadavid & Garcia, 2016; Porter, 2008). Organisational performance here focuses on the achievement of desired financial and non-financial performance outcomes. It is measured based on Kaynak and Kara's (2004) measure and captures the performance outcomes of sales, product quality and employee turnover. The focus on the relationship between the intensity of competitive forces and organisational performance is pertinent as Porter (2008) infers that the collective strength of the five forces drives performance, in particular the profitability, of organisations. Competitive advantage is defined as "the advantages that a business may have over another that are difficult to imitate" (Langfield et al., 2018, p. 15). An organisation achieves competitive advantage when they are more successful than their current and potential rivals (Schilke, 2014) and therefore, we measure competitive advantage using Schilke's (2014) measure, which focuses on an organisations' performance in respect to market share, earnings before interest and taxes, return on investment and return on sales relative to the industry average. We focus on the effect of the intensity of competitive forces on competitive advantage

as the five forces influence the “nature of competitive interaction within an industry” (Porter, 2008, p. 79).

In addition, given the importance of information systems in managing competitive environments (Barba-Sánchez et al., 2018), we consider the moderating role of the use of management accounting practices on the association between the intensity of competitive forces with both competitive advantage and organisational performance¹. Specifically, the second objective is to examine the moderating role of traditional management accounting and contemporary management accounting practices on the association between the intensity of competitive forces and the two organisational outcomes (organisational performance and competitive advantage). While both traditional and contemporary management accounting practices are important organisational information systems aimed at providing relevant information that adds value to organisations and aids effective decision-making, they are considered to be distinct (Langfield-Smith et al., 2018), categorised based on the period of their development and their features. Specifically, traditional management accounting practices are regarded as those management accounting practices such as budgeting and standard costing that were developed before the 1980s. They tend to be more short-term in focus and financially oriented (Dahal, 2021; Jaradat et al., 2021). Alternatively, contemporary management accounting practices are regarded as those management accounting practices popularised since the 1980s, with their distinctive features including being strategic in focus (i.e., linking processes, operations, and/ or activities with strategic outcomes), being both historical and future-oriented and inter-organisational in nature (Abdel-Kader & Luther, 2006; Dahal, 2020, 2021; Jaradat et al., 2021; Kaplan & Norton, 2001; Watts et al., 2014).

We contend that management accounting practices work as a package (Sandelin, 2008) and focus on the moderating role of the package of contemporary and traditional management accounting practices. Specifically, consistent with prior studies (i.e., Bedford et al., 2016; Chenhall, 2003 and Malmi & Brown, 2008), we envisage that management controls and these associated management accounting practices don't operate in a disjointed manner, but rather operate as a package and hence, we bundle the relevant practices together as higher order constructs, namely traditional and contemporary management accounting practices. Much of this literature suggests that bundling or packaging accounting practices together is appropriate as it recognizes the interlinkages between these relevant practices. Specifically, Malmi and Brown (2008, p. 287) state that ‘while much of the MCS research considers single themes or practices that are seemingly unconnected from each other and the context in which they operate, these invariably sit within a broader control system’. In addition, bundling the practices as packages alleviates the concern that the relations found between some contingency variables and individual control and accounting practices is weak and fragmented (Chenhall, 2003; Dent, 1990).

¹ We use the ordinal independent variable interaction approach (see Luft & Shields, 2003) whereby we consider both the direct effect of contemporary and traditional management accounting and the effect of their interaction with the intensity of competitive forces practices on competitive advantage and organisational performance.

As discussed in detail in the literature review section, our focus on the moderating role of the package of contemporary and traditional practices is considered pertinent, as the focus of such practices on cost management, cost control, quality, flexibility, innovation, and value creation (Langfield-Smith et al., 2018; Chenhall & Langfield-Smith, 1998a) has the potential to combat the industry's competitive forces. Therefore, using a survey distributed to the Financial Managers of 505 U.S. based organisations, as depicted in Fig. 1, the study provides an empirical insight into the association between the intensity of competitive forces with organisational outcomes (organisational performance and competitive advantage) and the moderating role of the package of management accounting practices (contemporary and traditional) on those associations.

The findings provide evidence of a positive association between the intensity of competitive forces with both organisational performance and competitive advantage. In addition, we find that the use of traditional management accounting practices positively moderates the association between the intensity of competitive forces and competitive advantage. Additional exploratory analysis further highlights the positive moderating role of the use of contemporary management accounting practices on the association between the intensity of competitive forces and organisational performance (competitive advantage) for firms employing a product differentiation (low cost) strategy and experiencing a high intensity of competitive forces. The additional exploratory analysis also shows that the use of contemporary (traditional) management accounting practices negatively moderates the association between the intensity of competitive forces and competitive advantage (organisational performance) in the product differentiation/low-force (high-force) model.

The findings contribute to Porter's (1979) competitive strategy literature by providing an empirical insight into the positive effect of the intensity of competitive forces on organisational outcomes (both organisational performance and competitive advantage). Furthermore, the findings contribute to management accounting

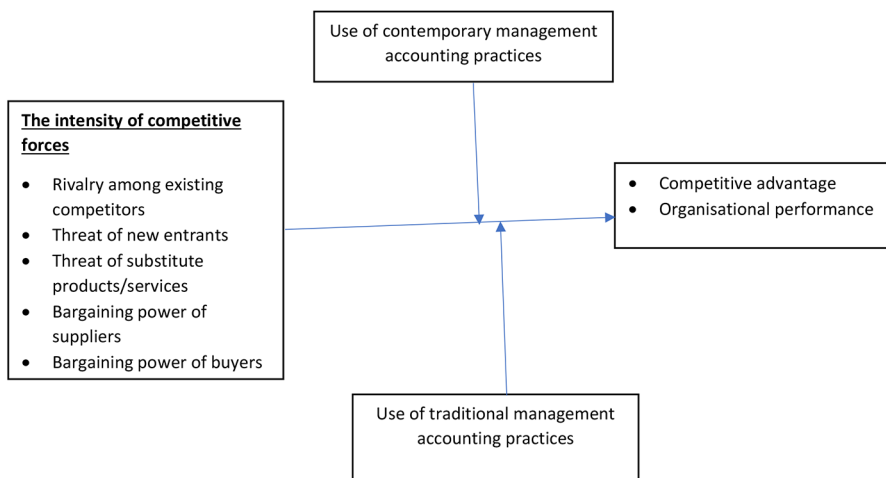


Fig. 1 The research model

research, in particular the emerging line of research on the strategic role of management accounting practices, by informing practitioners of the appropriate management accounting practices to deploy in different circumstances in order to enhance the effects of the intensity of competitive forces.

2 Literature review and hypotheses development

2.1 Porter's (1979) competitive forces

Porter (1979) developed a competitive forces model to provide a means for industries and organisations to analyse, evaluate, and react to their competitive environment by formulating appropriate strategies and taking appropriate actions to influence and/or mitigate the detrimental effects on their competitive environment. Each of the five forces (threat of new entrants, bargaining power of suppliers, bargaining power of buyers, rivalry among existing competitors, and threat of substitutes) is expected to influence organisational performance, specifically a measure of overall performance including profitability, product and employee performance, and competitive advantage, reflecting the relative performance compared to its competitors in areas of profitability and market share (Schilke, 2014).

The threat of new entrants “puts pressure on prices, costs, and the rate of investment required to compete” (Porter, 2008, p. 80). In particular, there is a greater need to maintain or reduce prices to maintain market share and deter potential future competitors, resulting in detrimental effects on profitability and competitiveness. Similarly, powerful suppliers and buyers can have detrimental impacts on profitability and competitiveness. With respect to suppliers, this can transpire as suppliers impose higher selling prices, limit quality or services, pass on costs, and/or integrate backward to compete in the market. With respect to buyers, powerful buyers “can capture more value by forcing down prices, demanding better quality or more service (thereby driving up costs)..... all at the expense of industry profitability” (Porter, 2008, p. 83). Greater rivalry or competition among organisations competing in an industry inhibits an organisation's ability to achieve a competitive advantage. At the same time, such competition can inhibit profitability as organisations attempt to gain an edge through price discounting or introducing new products, enhancing after-sales service, and/or advertising campaigns (Porter, 2008), all of which increase costs. Similarly, when there is a high threat of substitutes, organisations are forced to incur extra costs in an attempt to distinguish their products from others through product enhancements and/or marketing campaigns, while their ability to influence selling prices is limited.

Porter (1979) maintained that these five forces, characterising the overall industry structure, determine the attractiveness of an industry or sector. In particular, as all five forces are expected to have a detrimental effect on the profitability and competitiveness of an industry and individual organisations, it is expected that greater competitive intensity will limit the potential to obtain such benefits (Barba-Sánchez et al., 2018). Similarly, Porter (2008) maintains that the intensity of competitive forces makes it difficult to earn returns on investments. Therefore, we hypothesise that the

intensity of competitive forces is negatively associated with organisational performance and competitive advantage².

Hypothesis 1a The intensity of competitive forces is negatively associated with organisational performance.

Hypothesis 1b The intensity of competitive forces is negatively associated with competitive advantage.

2.2 The moderating role of management accounting practices

Management accounting practices are typically regarded as tools and techniques that facilitate planning, control, and decision-making (Langfield-Smith et al., 2018; Naranjo-Gil & Hartmann, 2006). Management accounting practices are broadly categorised as traditional management accounting practices (e.g. Standard Costing, Variance Analysis, and Return on Investment) which are considered more financially focused, narrower in scope, and more rigid; and contemporary management accounting practices (e.g. Activity-Based Costing / Activity-Based Management, Benchmarking, and the Balanced Scorecard) which are considered more inter-organisational in focus, multidimensional in nature, and cover a wider range of critical success factors (Chenhall & Langfield-Smith, 1998a, b).

While previous studies (for example, Grafton et al., 2010; Hoque & James, 2000; Kennedy & Affleck-Graves, 2001) have highlighted the important role of both traditional and contemporary practices in promoting organisational performance and competitive advantage, this study focuses on the moderating role of such practices in influencing the association between the intensity of competitive forces with both organisational performance and competitive advantage. In particular, we maintain that through providing information which enhances an organisation's understanding of their competitive forces, and which informs decision making in respect to the management of such forces, management accounting practices will positively moderate the association between the intensity of competitive forces with organisational performance and competitive advantage. This is supported by Porter (2008, p. 80) who indicates that "understanding the competitive forces, and their underlying causes, reveals the roots of an industry's current profitability while providing a framework for anticipating and influencing competition (and profitability) over time".

Hence, management accounting practices have the potential to assist in managing competitive forces through their decision support and information provision roles (Oboh & Ajibolade, 2017). However, there is sparse empirical research on the inter-relationships between management accounting practices and the effects of competitive pressures (i.e., competitive forces). Therefore, our study addresses this gap in the literature by focusing on the role of management accounting practices in coping with industry competitive pressures (i.e. the intensity of competitive forces). Specifically, we propose that the ability of organisations to cope with the intensity of competitive

² While the literature suggests a negative association, as the empirical examination of such associations is in its infancy, these hypotheses are somewhat exploratory.

pressures in a way which enhances organisational outcomes (i.e. organisational performance and competitive advantage), is influenced by the management accounting practices employed (Chenhall & Langfield-Smith, 1998a; Laela et al., 2018; Barbuto Jr, 2016).

2.2.1 The moderating role of contemporary management accounting practices

Although there is a dearth of qualitative and empirical studies on the role of contemporary management accounting practices in managing competitive forces, the potential role of such practices in managing competitive forces is apparent due to the underlying characteristics of these practices. In particular, through providing organisations with intelligence on their customers (Liu & Atuahene-Gima, 2018) and competitors (Kohli & Jaworski, 1990), contemporary management accounting practices assist in enabling organisations to manage the intensity of competitive forces. In particular, the external and flexible orientation (Stormi et al., 2019) of specific contemporary management accounting practices including Benchmarking, Value Chain Analysis, Activity-based Management and Activity Based Costing, will inform and enhance the effectiveness of managing competitive forces. For instance, through engaging in the process of comparing their performance against competitors (i.e. Benchmarking), managers will be more aware of the threats of substitute products and/or new entrants, and of their performance relative to competitors. Accordingly, they will be more informed and hence, in a better position to take appropriate action to improve their performance relative to their competitors. Further, Value Chain Analysis will assist managers in assessing the nature of the power of buyers and suppliers in their industry, while Activity-Based Costing and Activity-Based Management can be used to assess the specific activities and the cost of the activities involved with dealing with specific suppliers and buyers (i.e. customers) in their industry. Such information will enable managers to evaluate and reassess their competitive position in the value chain and better manage the impact of these competitive forces.

In addition, while contemporary management accounting practices are typically considered to be multidimensional, and quality and externally oriented, some contemporary practices, including Strategic Cost Management and Activity-Based Costing / Activity-Based Management, also focus on cost management, due to their “ability to identify sources of cost, manage and reduce costs, and eliminate wasteful activities” (Nuhu et al., 2016, p. 73). Further, contemporary management accounting such as the Balanced Scorecard, the use of Key Performance Indicators and Strategic Cost Management inform managers of their ability to meet internal performance targets in key performance areas. Accordingly, through evaluating their ability to meet performance targets, managers will be better informed of areas requiring improvement and better able to take action that enhances their competitive position and organisational performance. As such performance measures incorporate performance in relation to buyers (e.g. customer satisfaction measures) and suppliers (e.g. supplier performance indices), these contemporary management accounting practices will also facilitate better management of the buyer and supplier power competitive forces.

Finally, through adopting a continuous improvement philosophy, Total Quality Management will enable organisations to be more competitive, minimise the threat

of new entrants or substitute products, and hopefully tip the balance of power back in their hands as opposed to that of buyers. Specifically, as “Total quality management (TQM) is an integrative organizational-wide philosophy aimed towards continuously improving the quality of products/services and processes in order to meet or exceed customer expectations” (Baird et al., 2011, p. 789), it is expected to assist organisations in managing the competitive forces in a way which leads to a higher competitive advantage and organisational performance. In addition, there is evidence that quality data is associated with “identifying and solving supplier problems more efficiently” and “providing guidance on areas requiring correction” (Baird et al., 2011, p. 793). Such supplier quality management can assist in improving relationships with suppliers (Baird et al., 2011) and in enhancing process management and product / service design, thereby assisting in minimising both supplier and buyer power.

Therefore, we argue that contemporary management accounting practices will assist in enhancing an organisations’ ability to cope with the intensity of competitive forces in a way which can enhance both their competitiveness (i.e. competitive advantage) and profitability. Specifically, it is expected that there will be a stronger (weaker) association between the intensity of competitive forces with organisational outcomes (competitive advantage and organisational performance) when there is a higher (lower) extent of use of contemporary management accounting practices. Hence, we develop the following hypothesis:

Hypothesis 2 *The extent of use of contemporary management accounting practices positively moderates the association between the intensity of competitive forces and organisational outcomes (competitive advantage and organisational performance).*

2.2.2 The moderating role of traditional management accounting practices

Traditional management accounting practices were developed in the early 19th century during the industrial revolution and mainly focus on efficiency and cost control. The cost control focus of traditional management accounting techniques (Chenhall & Langfield-Smith, 1998a, p. 248) such as Budgeting, Cost / Benefit Analysis, Standard Costing and Variance Analysis should assist organisations in managing their day-to-day operational activities and through informing their tactical short-term decision making. Accordingly, the information provided by such traditional management accounting practices will facilitate an organisation’s management of their rivalry amongst existing competitors in a way which leads to enhanced competitive advantage and improved performance. Further, through providing managers with information in relation to the costs associated with generating customer value (Liu & Atuahene-Gima, 2018) and dealing with specific suppliers, traditional management accounting practices will also enable managers to make informed decisions in an attempt to minimise the power of suppliers and buyers, thereby enhancing their competitive position and profitability. Finally, traditional management accounting practices such as Capital budgeting and Return on Investment will assist organisations by providing them with information on the cost of alternative courses of action

(Langfied-Smith et al., 2018), thereby enabling managers to respond to and cope with the intensity of competitive forces more effectively.

Therefore, we argue that traditional management accounting practices will play a positive moderating role, enhancing an organisations' ability to cope with the intensity of competitive forces in a way which can enhance both their competitiveness (i.e. competitive advantage) and profitability. Specifically, it is expected that there will be a stronger (weaker) association between the intensity of competitive forces with organisational outcomes (competitive advantage and organisational performance) when there is a higher (lower) extent of use of traditional management accounting practices. Therefore, we hypothesise the following:

Hypothesis 3 *The extent of use of traditional management accounting practices positively moderates the association between the intensity of competitive forces and organisational outcomes (competitive advantage and organisational performance).*

3 Method

Data were collected using the survey research method which is commonly used in management accounting research (Baird et al., 2019; Chenhall, 2012; Nuhu et al., 2017; Su et al., 2017). This method enabled us to obtain data from a large sample of practitioners, thereby enhancing the generalisability of the results. The survey questionnaire was administered by Qualtrics, a data collection company with global outreach and experience in data collection management. A sample size of 505 respondents from the United States was obtained. Following a review of the U.S. Bureau of Labor Statistics' Standard Occupational Classification (SOC) system codes, employees classified as 11 3031 Financial Managers were chosen as the ideal respondents because of their knowledge of competitive forces, the use of management accounting practices, and organisational outcomes (competitive advantage and organisational performance). The questionnaire was targeted at large organisations, that is, those with 100 or more employees, due to the complexities associated with implementing management accounting practices and the commitment of resources required (Heggen, 2019) i.e. such practices are more likely to be utilised by larger organisations.

A total of 1600 potential respondents were contacted with 440 of those deemed ineligible to complete the questionnaire as they did not give consent (57), were not based in the U.S. (26), were not financial managers (132), or worked in organisations with less than 100 employees (225). Of the remaining 1160 contacts, 548 completed questionnaires were received, a response rate of 47.24%. However, as speeder checks removed 10 of these and a further 33 were removed due to repetitive responses, 505 complete responses were available for data analysis (i.e. a response rate of 43.53%). Table 1 shows that the majority of the responding organisations had 100–499 employees (52.9%), 135 (26.8%) had 500–999 employees, 78 (15.5%) had 1000–5000 employees, and only 24 (4.8%) organisations had over 5000 employees. As the usefulness of management accounting practices may vary across industries,

Table 1 Profile of responding organisations

Organisational Size (Number of employees)		
Between 100 and 499	266	52.9
Between 500 and 999	135	26.8
Between 1000 and 5000	78	15.5
Over 5000	24	4.8
Total	503	100%
Industry		
Wholesale trade	38	7.5
Manufacturing	64	12.7
Agriculture, forestry and fishing	9	1.8
Finance, insurance and real estate	187	37.0
Construction	34	6.7
Mining	16	3.2
Services	75	14.9
Public administration	21	4.2
Retail trade	32	6.3
Other	29	5.7
Total	505	100%

the questionnaire was distributed to organisations from various U.S. industries, and respondents were asked to identify their industries in accordance with the Standard Industrial Classification (SIC) primary codes. Table 1 shows that the majority of the responding organisations were from the ‘Finance, insurance and real estate’ industry (37%), followed by the ‘Services’ (14.9%) and ‘Manufacturing’ (12.7%) industries, with the remaining categories each representing less than 10% of the total responding organisations.

3.1 Common method bias

To address potential concerns of common method bias associated with the use of self-reported data, the following procedures were implemented. First, the questionnaire was pre-tested with academics with relevant research expertise to ensure the content validity and clarity of the measurement scales. Second, respondents were assured of the anonymity of their responses to minimise the effects of evaluative apprehension. Third, Harman’s (1967) test was employed, with the results indicating that the highest variance explained by a single factor was 35.5%, which was well below the recommended threshold of 50%. Fourth, Kock’s (2015) full collinearity test was conducted, with the results indicating that the variance inflation factors at the construct level ranged from 1 to 2.68 i.e. below the recommended threshold of 3.3. Finally, we externally validated the survey data to ensure validity. Specifically, given that some respondents (32) identified their entities, we obtained the most recently reported financial information, namely, return on assets (ROAs), from FactSet for these entities and correlated the data with the self-reported measures of organisational performance and competitiveness (i.e. competitive advantage) from the survey. We found that the externally reported ROA amounts were positively correlated with

both organisational performance (correlation coefficient=16.5%) and competitive advantage (correlation coefficient=11.5%)³. Hence, overall these procedures suggest that common method bias does not pose a serious concern.

3.2 Measurement of variables

3.2.1 Intensity of competitive forces

As previously discussed, most studies examining Porter's (1979) competitive forces have been qualitative; hence, there are limited studies which have empirically assessed the intensity of competitive forces (Barba-Sánchez et al., 2018; Marx, 2017). These studies assessed the intensity of competitive forces as the collective strength of five items, with each of the five forces assessed using single-item measures. For instance, using five-point Likert scales, rivalry amongst existing competitors was assessed from "very low" to "very high"; threat of new entrants was assessed from "very difficult to enter" to "very easy to enter"; threat of substitutes was assessed from a "very low threat" to a "very high threat"; and the bargaining power of suppliers and buyers was assessed as "very weak power" to "very strong power".

This study employs a similar approach whereby the intensity of competition is assessed as the collective strength of the five forces. However, to improve the reliability and validity of the measure and amidst calls to consider the multidimensionality of the construct (Carle et al., 2005), as alluded to by Porter (1979), we incorporated multiple items to assess each of the five forces. These items were adapted from Kulmia (2014) and developed based on Porter's (2008) discussion, to reflect the nature of each of the five competitive forces.

Given the multidimensionality of the construct, the intensity of competitive forces was initially operationalised as a second-order construct consisting of the five individual forces, with the loading factors reported in Appendix A and the fit statistics shown in Table 2. However, Table 2 shows that the intensity of competitive forces has poor convergent validity (i.e. AVE=0.445), and most of the individual five forces have poor reliability (i.e. cronbach's alpha and composite reliability scores less than 0.7) when assessed based on the second-order construct. Accordingly, in line with

Table 2 Second-order construct reliability and validity of intensity of competitive forces

	Cronbach's alpha	Composite reliability	Average variance extracted (AVE)
Intensity of competitive forces	0.922	0.923	0.445
Bargaining power of suppliers	0.790	0.733	0.544
Rivalry among existing competitors	0.644	0.649	0.584
Bargaining power of buyers	0.673	0.676	0.605
Threat of substitute products	0.633	0.636	0.731
Threat of new entrants	0.781	0.783	0.604

³ The correlation coefficients reflect the correlatedness of the externally reported ROA with our survey measures of organisational performance and competitive advantage. For instance, 16.5% of the organisational performance can be explained by the externally reported ROA.

Barba-Sánchez et al. (2018), we used an alternative approach, whereby the intensity of competitive forces was assessed based on the average score of the five forces.

We conducted an exploratory factor analysis (EFA) separately for each construct (dimension) of the intensity of competitive forces and the other variables. A single exploratory factor analysis for all of the items (i.e., 26 items) used to measure the five dimensions of the intensity of competitive forces was not conducted since a single factor analysis for a large number of items with high internal consistency may not yield the number of expected factors (Su et al., 2017) and given each dimension of the intensity of competitive forces is distinct (Mathooko & Ogutu, 2015). The EFA results revealed that the items used to measure each of the five forces loaded on the same dimension (see Appendix B for the list of the items). Confirmatory Factor Analysis (CFA) was then performed and resulted in the removal of some items (see Appendix B) due to low loadings. Next, we computed the score for each of the five forces as the average score of the items retained, with the intensity of competitive forces then calculated as the average score across these five forces, with higher (lower) scores representing a higher (lower) intensity of competitive forces⁴. Table 3 shows that this overall measure of the intensity of competitive forces was reliable with a Cronbach's alpha score of 0.907, a composite reliability score of 0.931, and an average variance extracted (AVE) score of 0.730, all of which exceeded the required thresholds (0.7, 0.7, and 0.5, respectively) (Chin, 1998; Nunnally, 1978; Werts et al., 1974), thereby providing support for the reliability and convergent validity of the measure⁵.

3.2.2 Organisational performance

Organisational performance was measured using a six-item measure (see Appendix B) based on Kaynak and Kara (2004). Respondents were required to indicate the extent to which they agreed that each of six performance related outcomes were achieved on a five-point Likert scale with anchors of "Strongly agree" and "Strongly disagree". Three items were removed due to low loadings with organisational performance measured as the average score of the remaining three items, with higher (lower) scores indicating higher (lower) organisational performance. While the Cronbach's alpha score of 0.636 is below Nunnally's (1978) cut-off of 0.7, the composite reliability score of 0.805 and AVE score of 0.579 provide support for the reliability and convergent validity of the measure.

⁴ While we acknowledge the approximative nature of this approach, given each of the 5 forces were measured using the same Likert scales, ranging from very low (1) to very high (5), it enables us to gain an empirical insight into the overall intensity of competitive forces. Furthermore, this approach is consistent with the empirical approach adopted by prior studies, for example, the measurement of sustainability as the average of the three dimensions of sustainability in Firmansyah et al. (2021).

⁵ We acknowledge that the mean score here is quite high and hence, our reference to an organisation with a high-force refers to those facing a very high intensity of competitive forces (4 or higher). At the same time, given this high mean score, those organisations classified as facing a low intensity of competitive forces (less than 3.99) could still face reasonably high levels of competitive forces.

Table 3 Descriptive statistics

	Mean	Standard deviation	Theoretical (actual) Minimum	Theoretical (actual) Maximum	Cronbach's Alpha	Composite Reliability	Average Variance Extracted
Intensity of competitive forces	3.99	0.63	1.00 (1.24)	5.00 (5.00)	0.907	0.931	0.730
Organisational performance	4.13	0.67	1.00 (1.00)	5.00 (5.00)	0.636	0.805	0.579
Competitive advantage	4.10	0.62	1.00 (1.00)	5.00 (5.00)	0.717	0.825	0.541
Contemporary MAPs	4.00	0.68	1.00 (1.50)	5.00 (5.00)	0.716	0.824	0.540
Traditional MAPs	4.07	0.61	1.00 (1.25)	5.00 (5.00)	0.693	0.812	0.520
Differentiation strategy	4.02	0.65	1.00 (1.00)	5.00 (5.00)	0.710	0.821	0.534
Low-cost strategy	4.06	0.67	1.00 (1.00)	5.00 (5.00)	0.639	0.806	0.581

3.2.3 Competitive advantage

Competitive advantage was measured using Schilke's (2014) six-item measure (see Appendix B). Two of these items were removed because of low loadings with the remaining four items reporting a Cronbach's alpha score of 0.717, a composite reliability score of 0.825, and an AVE of 0.541, all of which exceeded the required cut-offs, thereby supporting the reliability and convergent validity of the measure. Hence, competitive advantage was measured as the average score of the four items, with higher (lower) scores indicating greater (weaker) competitive advantage.

3.2.4 The extent of use of management accounting practices

The extent of use of traditional and contemporary management accounting practices was measured using the list of practices identified by Chenhall and Langfield-Smith (1998b). While there have been new innovations in management accounting, in line with prior studies that have assessed the use of management accounting practices based on Chenhall and Langfield-Smith's (1998b) due to its comprehensiveness (e.g. Agbejule & Huusko, 2011; Angelakis et al., 2010; Dahal, 2021), this study also assesses the extent of use of traditional and contemporary management accounting practices based on Chenhall and Langfield-Smith (1998b).

Specifically, the extent of use of traditional management accounting practices was assessed based on the extent of use of six traditional practices (see Appendix B), using anchors of "Not at all" and "To a great extent". Two of these were removed due to low loadings with the extent of use of the package of traditional management accounting practices measured as the average score of the extent of use of the remaining four practices. The measure reported a Cronbach's alpha of 0.716, composite reliability score of 0.824, and an AVE of 0.540, all above the required cut-off values. Hence, the reliability and convergent validity of the constructs were ensured.

The extent of use of contemporary management accounting practices was assessed in respect to the extent of use of use of eight contemporary management accounting practices (see Appendix B) using anchors of "Not at all" and "To a great extent". Four of these were removed due to low loadings with the extent of use of the package of contemporary management accounting practices measured as the average score of the extent of use of the remaining four practices. While the Cronbach's alpha score (0.693) was just below the 0.7 cut-off, the composite reliability score (0.812) and the AVE score (0.520) exceeded the required cutoff; hence, there was strong support for the reliability and convergent validity of this construct.

3.2.5 Competitive strategy

Additional exploratory analysis was undertaken to examine the hypothesised relationships for organisations employing different strategies. For this purpose, Zahra and Covin's (1993) constructs were used to assess the competitive strategy of organisations. Zahra and Covin (1993) used a seven-item measure of product differentiation strategy and a five-item measure of low-cost strategy (see Appendix B). With respect to the measure of the product-differentiation strategy, three items were removed due

to low loadings with the focus on the product-differentiation strategy measured as the average score of the remaining four items. With respect to the measure of the low-cost strategy, two items were removed due to low loadings, with the focus on the low-cost strategy measured as the average score of the remaining three items.

While the Cronbach's alpha score for the low-cost strategy measure was below the 0.7 cut-off (Nunnally, 1978), the composite reliability (0.806) and AVE scores (0.581) exceeded the cutoff values indicative of a reliable measure, whereas all three measures exceeded these cutoff values with respect to the product-differentiation strategy measure (0.710, 0.821, and 0.534, respectively). Hence, there is strong support for the reliability and convergent validity of these measures.

4 Results

Table 3 provides the descriptive statistics and measures of reliability for the constructs. The mean score for the intensity of competitive forces was quite high (3.99), as were the levels of organisational performance (4.13) and competitive advantage (4.10). The mean score for the extent of use of traditional management accounting practices (4.07) is slightly higher than the extent of use of contemporary management accounting practices (4.00), while the mean score for the use of a low-cost strategy is slightly higher (4.06) than the mean score for the product-differentiation strategy (4.02).

As discussed in the previous section, Table 3 highlights that the Cronbach's alpha scores exceed Nunnally's (1978) cut-off (0.7) for four of the seven constructs, with the exceptions being in relation to organisational performance (0.636), traditional management accounting practices (0.693) and the low-cost strategy (0.639). However, Table 3 shows that the composite reliability and AVE scores for these constructs are well in excess of the 0.7 and 0.5 required cut-offs, thereby assuring the reliability and convergent validity of these measures. Furthermore, in addition to a Cronbach's alpha score above 0.7, the composite reliability and AVE scores for the other four constructs were satisfactory, thereby providing strong overall support for the reliability and convergent validity of the study's constructs.

Table 4 provides the correlations between the constructs and the square root of the AVE scores for each construct. The square root of the AVE scores exceed the correlations with the other constructs for 20 out of the 21 correlations with the only exception being a slightly higher correlation between the extent of use of traditional management accounting practices and the differentiation strategy. As we do not hypothesise any relationship between these two variables this is not expected to be a concern and hence, we conclude that there are no problems in respect to the discriminant validity of the constructs.

4.1 Structural equation modelling

Structural equation modelling was used to examine the hypotheses. The results of the SEM analysis are shown in the initial 'overall sample' column of Table 5. This

Table 4 Correlations and square root of AVE scores

	Intensity of competitive forces	Organisational performance	Competitive advantage	Contemporary MAPs	Traditional MAPs	Differentiation strategy	Low-cost strategy
Intensity of competitive forces	0.854						
Organisational performance	0.543	0.761					
Competitive advantage	0.577	0.681	0.736				
Contemporary MAPs	0.618	0.576	0.630	0.735			
Traditional MAPs	0.622	0.569	0.605	0.685	0.721		
Differentiation strategy	0.676	0.535	0.604	0.692	0.727	0.731	
Low-cost strategy	0.663	0.507	0.563	0.641	0.703	0.729	0.762

NB The diagonal figures in bold represent the square root of the average variance extracted scores for each construct

Table 5 Structural equation modelling results

	Overall sample Path coeff. (<i>P</i> sig.)	Low-cost High-force Path coeff. (<i>P</i> sig.)	Low-cost Path coeff. (<i>P</i> sig.)	Product differentiation High-force Path coeff. (<i>P</i> sig.)	Product differentiation Low-force Path coeff. (<i>P</i> sig.)
Intensity of competitive forces → Organisational performance	0.244 (0.000)**	0.624 (0.012)*	0.056 (0.637)	0.723 (0.002)**	0.306 (0.083)
Intensity of competitive forces → competitive advantage	0.210 (0.000)**	0.094 (0.624)	0.193 (0.074)	0.702 (0.000)**	-0.123 (0.244)
Contemporary management accounting practices → Organisational performance	0.272 (0.000)**	0.182 (0.237)	0.388 (0.015)*	-0.141 (0.422)	0.306 (0.083)
Contemporary management accounting practices → Competitive advantage	0.284 (0.000)**	-0.115 (0.335)	0.204 (0.158)	0.137 (0.333)	0.117 (0.083)
Traditional management accounting practices → Organisational performance	0.305 (0.000)**	0.544 (0.010)*	0.207 (0.237)	0.817 (0.000)**	0.084 (0.461)
Traditional management accounting practices → Competitive advantage	0.284 (0.000)**	0.112 (0.495)	0.451 (0.004)**	0.343 (0.051)	0.301 (0.094)
Intensity of competitive forces x Contemporary management accounting practices → Organisational performance	0.044 (0.553)	0.123 (0.754)	0.152 (0.453)	0.978 (0.043)*	0.080 (0.695)
Intensity of competitive forces x Contemporary management accounting practices → Competitive advantage	-0.120 (0.065)	0.784 (0.010)*	-0.063 (0.733)	0.239 (0.540)	-0.428 (0.020)*
Intensity of competitive forces x Traditional management accounting practices → Organisational performance	0.082 (0.306)	-0.987 (0.094)	-0.040 (0.844)	-1.360 (0.009)**	-0.143 (0.589)
Intensity of competitive forces x Traditional management accounting practices → Competitive advantage	0.154 (0.028)*	0.189 (0.679)	0.147 (0.428)	-0.501 (0.236)	0.181 (0.446)
Goodness of fit					
CMIN/DF	1.023	0.343	1.380	0.042	1.318
GFI	0.999	0.999	0.992	1.000	0.993
AGFI	0.982	0.978	0.864	0.998	0.867
CFI	1.000	1.000	0.997	1.000	0.998
RMSEA	0.007	0.000	0.065	0.000	0.060

*significant at 5%

**significant at 1%

model exhibits strong goodness-of-fit scores⁶ and reveals that the intensity of competitive forces is significantly positively associated with organisational performance ($\beta=0.244, p=0.000$) and competitive advantage ($\beta=0.210, p=0.000$). Such associations are in the opposite direction to our hypotheses and hence, H1a and H1b are rejected.

The unexpected positive associations suggest that the threats posed by the five forces serve to focus managers' attention on the reconfiguration and recombination of their firms' resources and capabilities in a way which neutralizes the effects of these forces, thereby facilitating the achievement of competitive advantage and higher organisational performance. Past research here provides support for the notion that management proactively and consciously make decisions which alleviate the pressures from the competitive forces in a way which achieves higher performance. For example, managers and firms develop specific market orientations, i.e. processes of generating market intelligence to make sense of the key stakeholders in firms' marketplace, such as customers and competitors (Crick, 2021; Kohli & Jaworski, 1990; Narver & Slater, 1990), to counter the threats from the competitive forces. Market orientations have also been found to influence innovation through uncovering customer intelligence and then using it to generate superior customer values (Liu & Atuahene-Gima, 2018; Narver & Slater, 1990). In addition, market orientations facilitate greater intelligence on firms' competitors, thereby informing the development of strategies to remain competitive (Kohli & Jaworski, 1990). For example, Han et al. (1998) refer to how the analysis of competitors' product offerings led to a new competitive strategy involving the redevelopment of firm's product features, and subsequently a better product performance.

In respect to the hypothesised moderating effects, the initial 'overall sample' model in Table 5 shows that the use of contemporary management accounting practices did not moderate the effect of the intensity of competitive forces on organisational performance or competitive advantage and hence, H2 is rejected. However, Table 5 shows that the use of traditional management accounting positively moderates the association between the intensity of competitive forces and competitive advantage ($\beta=0.154, p=0.028$) and hence, partial support is provided for H3.

Finally, while not hypothesised, the model found that the two moderators were significantly positively associated with both organisational performance and competitive advantage. Specifically, the use of contemporary management accounting practices is significantly positively associated with organisational performance ($\beta=0.272, p=0.000$) and competitive advantage ($\beta=0.284, p=0.000$) and the use of traditional management accounting practices is significantly positively associated with organisational performance ($\beta=0.305, p=0.000$) and competitive advantage ($\beta=0.284, p=0.000$).

⁶ The recommended threshold guidelines are CMIN/DF<5; GFI>0.90; CFI>0.80; AGFI>0.80; RMS.EA<0.08 (Hair et al., 2010).

4.1.1 Additional exploratory analysis

Additional exploratory analysis was undertaken to examine the hypothesised relationships for organisations adopting different strategies and experiencing different levels of intensity of competitive forces. In respect to the intensity of competitive forces, we divided the respondents into two categories: those who experienced high (low) intensity of competitive forces based on whether their intensity of competitive forces was above (below) the mean score (3.99) for this variable. In respect to strategy, we compared the mean scores for the low-cost and product differentiation strategy measures, classifying organisations as predominantly focusing on the specific strategy that they reported the highest mean score for; that is, if the highest (lowest) mean score was reported with respect to the low-cost strategy, then they were classified as employing a low-cost (product-differentiation) strategy. We did however exclude respondents from the sample if the score for both strategies was below the mid-point of the range (i.e. less than 3), as such organisations failed to indicate a significant focus on either strategy.

The sample was divided into four quadrants (see Table 6), classified based on the strategy and strength of the forces i.e. competitive strategies (low-cost strategy vs. product differentiation strategy) versus intensity of competitive force (high vs. low). A comparison of the mean scores for each organisation revealed that 267 of the 505 (52.9%) organisations reported a higher score with respect to the product-differentiation strategy, with the remaining organisations (238, 47.1%) reporting a higher mean score for the low-cost strategy. However, as scores below the mid-point of the range (i.e. 3) indicate a weak emphasis on the dominant strategy, we removed these respondents leaving 252 (232) respondents for the product differentiation (low-cost) strategies respectively. The majority of these organisations reported an intensity of competitive forces score above the mean score of 3.99 (306), that is, a high force. Hence, Table 6 shows that out of the 252 organisations employing the product-differentiation strategy, 164 were subject to a high force and 88 were subject to a low force, while out of the 232 organisations employing a low-cost strategy, 142 (90) were subject to a high (low) intensity of competitive force.

Separate models were examined for respondents in each of the four quadrants with the results provided in Table 5⁷. Each of the additional models exhibit strong

Table 6 Classification of organisations into quadrants based on strength of force and strategy

Force ^a	High-force	Low-force
Strategy ^b		
Low-cost	142	90
Product-differentiation	164	88

^a Based on above (high) or below (low) the mean score of 3.99

^b While it is acknowledged that organisations may employ a hybrid strategy, they were categorised based on the highest mean score across the two strategies provided that the highest score exceeded the mid-point of the range (3) i.e. when the strategy score was 3 or less this was not deemed to be a clear strategy and they were removed

⁷ As this analysis was exploratory in nature, we aimed to examine the hypothesized relationship for each of the sub-groups (based on the classification of organisations into the four quadrants) separately. Future studies may choose to further analyse and compare the results for these sub-groups and/or additional categorisations of organisations.

goodness-of-fit scores and provide additional unique insights into the hypothesised relationships. For instance, additional support is provided for the rejection of H1a in the low-cost/high-force and product differentiation/high-force models, where the intensity of competitive forces is also significantly positively associated with organisational performance [$(\beta=0.624, p=0.012)$; $(\beta=0.723, p=0.002)$]. Further support is also provided for the rejection of H1b in the product differentiation/high-force model where the intensity of competitive forces is significantly positively associated with competitive advantage $(\beta=0.702, p=0.000)$.

In respect to the direct associations between the management accounting practices and the organisational outcomes, the significant positive association between the extent of use of contemporary management accounting practices and organisational performance is also found in the low-cost/low-force model, while further support is provided for the positive associations between the extent of use of traditional management accounting practices and organisational performance [competitive advantage] in the low-cost/high-force and product differentiation/high-force [low-cost/low-force] models.

In respect to the hypothesised moderating relationships, while there was no support for H2 in the overall sample model, there is evidence that the use of contemporary management accounting practices positively moderates the association between the intensity of competitive forces and organisational performance in the product-differentiation/high-force $(\beta=0.978, p=0.043)$ model and between the intensity of competitive forces and competitive advantage in the low-cost/high force model $(\beta=0.784, p=0.010)$. However, while such findings provide support for H2, in the product-differentiation/low-force model, the extent of use of contemporary management accounting practices exhibits a significantly negative moderating effect in the association between the intensity of competitive forces and competitive advantage $(\beta = -0.428, p=0.020)$.

Finally, while the overall sample model provides support for H3, with the extent of use of traditional management accounting practices found to positively moderate the effect of the intensity of competitive forces on competitive advantage $(\beta=0.154, p=0.028)$, there was only one significant finding in the four quadrant models, with the extent of use of traditional management accounting practices exhibiting a negative significant moderating effect on the association between the intensity of competitive forces and organisational performance in the product differentiation/high-force model $(\beta = -1.360, p=0.009)$. This finding suggests that the use of traditional management accounting practices is inappropriate when organisations employ a product differentiation strategy and the intensity of competitive forces is high.

5 Discussion and conclusion

This study sought to provide empirical insights into the effect of the intensity of competitive forces on the two organisational outcomes commonly associated with Porter's (1979) competitive forces, organisational performance and competitive advantage (Marin-Cadavid & Garcia, 2016; Porter, 2008). The findings highlight the positive effect of the intensity of competitive forces on organisational outcomes.

Specifically, the intensity of competitive forces exhibits a positive association with organisational performance and competitive advantage in the overall sample model, with organisational performance in the low-cost/high-force model and with both organisational performance and competitive advantage in the product differentiation/high-force models.

Such findings are in line with the literature which maintains that greater competitive intensity focuses manager's attention on using their resources in a way which alleviates the pressures from the forces and enables them to achieve higher competitive advantage and performance (Han et al., 1998; Kohli & Jaworski, 1990). Future studies can examine this relationship further to focus on the potential for organisations to utilise competitive forces in a way which enhances their competitive advantage and/or organisational performance and to explore the manner in which such benefits transpire.

Additionally, this study provides a preliminary empirical insight into the moderating role of the extent of use of contemporary and traditional management accounting practices on the association between the intensity of competitive forces and organisational outcomes (organisational performance and competitive advantage). The results in respect to the overall sample revealed that the extent of use of traditional management accounting practices (i.e. Capital Budgeting, Standard Costing, Variance Analysis and Return on Investment) was found to positively moderate the effect of the intensity of competitive forces on competitive advantage. In addition, the exploratory analysis revealed that the moderating effect of the use of contemporary and traditional management accounting practices differed based on the level of intensity experienced and the strategy employed. Specifically, the extent of use of the package of contemporary management accounting practices (i.e. Activity-Based Management, Balanced Scorecard, Value Chain Analysis, and Strategic Cost Management) was found to positively moderate the effect of the intensity of competitive forces on organisational performance (competitive advantage) in a product differentiation high-force (low-cost/high-force) environment. In addition, the extent of use of contemporary management accounting practices negatively moderated the association between the intensity of competitive forces and competitive advantage in the product differentiation / low-force model. Alternatively, while the extent of the use of traditional management accounting practices (i.e. Capital Budgeting, Standard Costing, Variance Analysis and Return on Investment) positively moderated the effect of the intensity of competitive forces on competitive advantage in the overall model, it was found to negatively moderate the effect of the intensity of competitive forces on organisational performance in the product-differentiation/high-force model.

These findings contribute to the literature by providing an empirical insight into the theoretical role of a specific internal mechanism, the extent of use of management accounting practices, in managing the effects of competitive forces within an organisation. Specifically, the findings highlight the important role of both contemporary and traditional management accounting practices in managing and/or mitigating the effect of the intensity of competitive forces on organisational performance and competitive advantage. From a practical perspective, the findings inform managers regarding the ideal management accounting practices to implement to manage competitive forces based on their prevailing strategy (low-cost or product differentiation)

and the intensity of the competitive forces they face (low or high). For instance, for managers whose organisations utilise a low-cost strategy, it is recommended that they use contemporary management accounting practices if the forces are more intense (i.e. high force), with the extent of use of the package of four contemporary management accounting practices exhibiting a positive moderating role on competitive advantage.

Managers whose organisations utilise a product-differentiation strategy should employ contemporary management accounting practices to a greater extent when the competitive forces are more intense (i.e. high-force), with such forces exhibiting a positive moderating effect on organisational performance. At the same time, they should not be focusing on using traditional management accounting practices given the negative moderating effect of such practices on the association between the intensity of competitive forces and organisational performance. In particular, this result implies that the focus on traditional rigid cost-focused measures (Cadez & Guilding, 2012; Nimtrakoon & Tayles, 2015) will have a detrimental impact on organisational performance by limiting the ability of organisations to cope with intense competitive forces in a way which enhances their organisational performance through product differentiation. If competitive forces are less intense (i.e. low force), then it is inappropriate to use contemporary management accounting practices to a greater extent, given that such practices negatively moderate the effect of competitive forces on competitive advantage. Hence, this result implies that contemporary management accounting practices inhibit organisations' ability to achieve competitive advantage through product differentiation when the intensity of competitive forces is low.

Overall, this study makes a significant contribution to the literature on competitive forces and management accounting by highlighting the important role of management accounting practices in coping with industry competitive pressures to enhance competitive advantage and organisational performance. In particular, our findings regarding the positive moderating role of both contemporary and traditional management accounting practices highlight the positive impact of such practices in effectively coping with the intensity of competitive pressures under specific circumstances based on the unique environment that organisations face, i.e. the combination of their prevailing strategy (low-cost or product-differentiation) and the intensity of the competitive forces (high or low) they face. Such findings provide an important initial empirical insight into the role of internal control mechanisms in coping with competitive forces, thereby highlighting the potential for future research to consider the role of alternative internal mechanisms, such as formal and informal control practices, in moderating the association between competitive forces and organisational outcomes.

This study is subject to the usual limitations associated with the survey method, including the inability to assert causation and common method bias. However, as previously discussed, we have undertaken a series of tests to alleviate these common method bias concerns, including Harman's (1976) and Kock's (2015) full collinearity tests, which serve to alleviate any concerns regarding common method bias. Further, as we have developed a new measure to assess each of Porter's (1979) five competitive forces and subsequently measure the intensity of competitive forces, future studies may look to confirm the reliability and validity of this measure. However, while this measure is new, it represents a more rigorous and intuitive approach to measur-

ing the intensity of competitive forces. In particular, as the measure encompasses the specific attributes of each force described by Porter (2018), it provides a more comprehensive account of the effects of the intensity of competitive forces to be used in future studies. In addition, while the study examined the moderating role of the management accounting practices in four quadrants it is acknowledged that the classification scheme employed to determine the strategy and intensity of competitive forces is open to interpretation, especially in relation to the intensity of competitive forces where the mean score was quite high possibly due to the data being collected during COVID-19 with even those organisations classified as being subject to a low-force experiencing quite high competitive pressures. Hence, while our additional analysis here was somewhat exploratory, future studies may continue to explore the effects of the intensity of competitive forces and the moderating role of management accounting practices on organisational outcomes in different circumstances. Finally, given the majority of the respondents are from the finance, insurance and real estate industry, this may limit the generalisability of the results. Accordingly, future studies may examine the hypothesised relationships in specific industries.

Appendix A: first-order and second-order measurement of intensity of competitive forces

Only retained items marked with a symbol (†) are reported, with the values indicating the standardised factor loadings of the retained items.

	First-order loading	Second-order loading
<i>Bargaining power of suppliers</i>		0.952
A small number of potential input suppliers can access goods and services from†.	0.599	
The goods and services purchased by our organisation are not widely available†.	0.770	
It is easy for our suppliers to become direct competitors by selling directly to our customers.	0.769	
It is difficult for our organisation to switch to substitute products from other supplier†.	0.602	
Our organisation has limited knowledge about our supplier's products and market†.	0.752	
<i>Rivalry among existing competitors</i>		1.032
There are no clear leaders in our market†.	0.753	
It is easy to compete in the industry because of low fixed costs†.	0.637	
Competition is high, as our products (goods and/or services) are standardised†	0.576	
<i>Bargaining power of buyers</i>		1.073
We have a limited number of customers whose purchases represent a large proportion of our total revenue from goods and/or services†.	0.661	
It would be easy for customers to integrate backward in the supply chain, that is, source goods and/or services from our suppliers†.	0.620	

	First-order loading	Second- order loading
It is easy for customers to switch to access goods and/or services from competitors†.	0.674	
<i>Threat of substitute products</i>		0.762
Substitute products compare favourably to our products (goods and/or services) on a price-performance basis†.	0.752	
Our customers are amenable to purchasing products (goods and/or services) from competitors†.	0.721	
<i>Threat of new entrants</i>		0.953
Low initial capital investments are required to commence new operations in the industry†.	0.687	
No specialised knowledge and/or technical expertise is required to commence operations in our industry†.	0.784	
New entrants can easily acquire/obtain customers in our industry†.	0.597	
New entrants can easily acquire and obtain the inputs required to compete efficiently in our industry†.	0.684	

Appendix B: measurement of constructs

Items marked with a symbol (†) were retained after testing the measurement models (i.e. CFA). The values in parentheses for each retained item indicate standardised factor loadings.

Intensity of competitive forces

Bargaining power of suppliers

A small number of potential input suppliers can access goods and services from† (0.681).

The goods and services purchased by our organisation are not widely available† (0.766).

Purchases from our suppliers represent a large portion of the suppliers' business.

It is easy for our suppliers to become direct competitors by selling directly to our customers. † (0.763).

It is difficult for our organisation to switch to substitute products from other suppliers† (0.671).

Our organisation has limited knowledge about our supplier's products and market† (0.792).

Rivalry among existing competitors

There is a large number of competitors in our industry.

There are no clear leaders in our market† (0.777).

The market is growing quickly.

It is easy to compete in the industry because of low fixed costs† (0.772).

Competition is high, as our products (goods and/or services) are standardised† (0.743).

Bargaining power of buyers

We have only a few customers, and hence, losing one is critical to our success.

We have a limited number of customers whose purchases represent a large proportion of our total revenue from goods and/or services† (0.820).

Our customers have a detailed knowledge of the range of goods and services available in the industry.

Customers can easily gain access to an equivalent product (goods and/or services).

It would be easy for customers to integrate backward in the supply chain, that is, source goods and/or services from our suppliers† (0.806).

It is easy for customers to switch to access goods and/or services from competitors† (0.700).

Threat of substitute products

Substitute products compare favourably to our products (goods and/or services) on a price-performance basis† (0.915).

Product diversity in our industry is low.

Our customers are amenable to purchasing products (goods and/or services) from competitors† (0.781).

Threat of new entrants

There are limited incumbent advantages in our industry.

Low initial capital investments are required to commence new operations in the industry† (0.802).

The assets needed to run our organisation's operations are easily accessible.

No specialised knowledge and/or technical expertise is required to commence operations in our industry† (0.763).

New entrants can easily acquire/obtain customers in our industry† (0.734).

New entrants can easily acquire and obtain the inputs required to compete efficiently in our industry† (0.807).

Organisational performance

Profit goals have been achieved.

Sales goals have been achieved† (0.765).

Return on investment goals have been achieved.

Our product(s) are of higher quality than those of our competitors† (0.740).

We have a higher customer retention rate than our competitors.

We have a lower employee turnover rate than our competitors† (0.776).

Competitive advantage

We have gained strategic advantages over our competitors.

We have a large market share† (0.756).

Overall, we are more successful than our major competitors.

Our earnings before interest and taxes (EBIT) continuously exceed the industry average† (0.744).

Our return on investment (ROI) is continuously above industry average† (0.657).

Our return on sales (ROS) is continuously above industry average† (0.780).

Contemporary management accounting practices

Benchmarking (e.g. quality, cost practices and procedures).

Activity Based Management† (0.769).

Activity Based Costing.

The Balanced Scorecard† (0.759).

Value Chain Analysis† (0.708).

Total Quality Management.

Key Performance Indicators.

Strategic Cost Management† (0.699).

Traditional management accounting practices

Budgeting for planning and control.

Capital budgeting† (0.672).

Cost benefit Analysis.

Standard Costing† (0.734).

Variance Analysis† (0.687).

Return on investment† (0.788).

Low-cost strategy

Efficiency in securing raw materials or components† (0.780).

Finding ways to reduce costs.
 The level of operating efficiency.
 The level of production capacity utilization† (0.768).
 Price competition† (0.738).

Product-differentiation strategy

Using new methods and technologies to create superior products† (0.672).
 New product development† (0.724).
 The rate of new product introduction to the market.
 The number of new products offered to the market† (0.761).
 Intensity of advertising and marketing.
 Developing and utilizing the sales force† (0.763).
 Building a strong brand identification.

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Data availability The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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